

Quick Activity 1: School Supplies Calculator

A notebook costs ₱45, and a pen cost ₱12. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the **total cost** and **change**.

Problem Analysis

Input	Process	Output
Number of notebook, number of pen, payment	Calculate notebook cost, calculate pen cost, add for total cost, calculate change	Total cost and change

Decomposition

1. Read the number of notebook.
2. Read the number of pen.
3. Read the payment amount.
4. Calculate the notebook cost.
5. Calculate the pen cost.
6. Calculate the total cost.
7. Calculate the change.
8. Display the total cost and change.

Pseudocode

START

```
READ notebook, pen, payment
notebookCost ← notebook x 45
penCost ← pen × 12
totalCost ← notebookCost + penCost
change ← payment – totalCost
DISPLAY totalCost
DISPLAY change
```

END

Dry Run Calculations

Test Case 1

Input:

- Notebook: 3
- Pen = 2
- Payments = ₱200

Step	Variable	Calculation	Solution
1	Notebook Cost	3×45	₱135
2	Pen Cost	2×12	₱24
3	Total Cost	$135 + 24$	₱159
4	Change	$200 - 159$	₱41

Output

Total Cost = ₱159

Change = ₱41

Test Case 2

Input:

- Notebook = 2
- Pen = 3
- Payment = ₱150

Step	Variable	Calculation	Solution
1	Notebook Cost	2×45	₱90
2	Pen Cost	3×12	₱36
3	Total Cost	$90 + 36$	₱126
4	Change	$150 - 126$	₱24

Output

Total Cost = ₱126

Change = ₱24